

CompTIA A+ — AP-A1

Hardware Components & System Assembly — Answer Sheet

Based on the supplied assessment file. Practical tasks should be completed on the actual hardware available and supported with your own photos/screenshots.

Section A — Concept Application

Question 1

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

Answer: First select the CPU according to the workstation requirements. Then choose a motherboard with the correct CPU socket and chipset support. Check the motherboard manual or manufacturer specifications for supported CPU families and BIOS requirements. Select RAM that matches the motherboard's supported memory type, speed range, capacity and slot configuration. Finally select storage that matches an available interface such as SATA or an M.2 slot with the required storage protocol. Before assembly, cross-check the manufacturer specifications for socket, chipset, RAM type, storage interface, physical fit, power requirements and BIOS support. After installation, enter BIOS/UEFI to confirm that the CPU, RAM and storage are detected correctly and then verify successful POST.

Question 2

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons.

Answer: HDDs store data on spinning magnetic platters and use mechanical read/write heads, so they are generally slower and more sensitive to mechanical shock. SSDs use flash memory and have no moving parts, giving faster access, lower latency and better durability. NVMe is a storage protocol designed for high-speed SSDs, normally using PCIe, and can provide much higher throughput and lower latency than SATA-based storage. For an office/productivity workstation, an SSD is the most suitable choice because it provides fast booting, application loading and file access without the mechanical limitations of an HDD. If the system supports it and the budget permits, an NVMe SSD is a strong choice for the primary drive.

Question 3

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under sustained graphics load.

Answer: The GPU processes graphics and parallel workloads for design applications. During sustained use it produces heat, so adequate cooling is necessary to keep temperatures within safe operating limits. The CPU cooler, GPU cooler, case fans and proper airflow help move heat away from components. Good thermal management prevents excessive temperatures, thermal throttling, crashes and reduced component life, allowing the workstation to maintain more consistent performance.

Question 4

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues from incompatible components.

Answer: Compatibility ensures that components can physically connect, communicate correctly and receive the required power. Problems can occur when a CPU does not match the motherboard socket or BIOS support, RAM is the wrong memory type, a storage device has no compatible interface or slot, the PSU lacks suitable connectors or sufficient capacity, or components do not physically fit the case. Incompatibility may result in failure to POST, components not being detected, instability, overheating or inability to install the hardware.

Question 5

Question: Explain how you would choose an appropriately rated PSU and describe safety and connection checks before powering on.

Answer: Estimate the power requirements of the CPU, GPU, storage, fans and other components, then select a reputable PSU with sufficient continuous wattage and appropriate connectors, with reasonable headroom for the system. Before powering on, switch off and unplug the PSU, inspect cables and connectors, and use ESD precautions. Confirm the motherboard main power connector, CPU power connector, GPU power connector where required, and storage power/data connections are fully seated. Check that no loose screws or tools are inside the case and that the cooler is correctly mounted. Then connect AC power and perform the first boot.

Question 6

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

Answer: On power-up, check that fans and indicators behave normally and that the system reaches the BIOS/UEFI screen without error messages or diagnostic beeps/LED codes. In BIOS/UEFI, confirm the CPU, installed RAM and storage device are detected. If POST fails, switch off power and troubleshoot systematically: check PSU connections, CPU power, RAM seating and configuration, GPU seating/power, motherboard indicators and storage connections. Reseat one component at a time and consult the motherboard diagnostic codes/manual. After correcting the likely fault, boot again and confirm successful POST.

Section B — Practical Coding Tasks

The assessment requires physical identification, removal/reinstallation and assembly. Use your own hardware observations for make/model/specifications and attach your own labeled photos or diagrams.

Task 1 — Component Identification & Functions

Component	Function / key points
Motherboard	Main circuit board connecting the CPU, RAM, storage, expansion cards, power and peripherals.
CPU	Executes instructions and performs general-purpose processing for the operating system and applications.
RAM	Temporary, high-speed working memory used by active programs and the operating system.
GPU	Processes graphics and other supported parallel workloads and outputs video.
Storage device	Provides non-volatile storage for the operating system, applications and files; may be SATA or NVMe.
Cooling system	Removes heat from the CPU/GPU and case to maintain safe operating temperatures and stable performance.

Photo/documentation: Label each physical component and record its visible make/model or key specification. Do not invent specifications that are not visible.

Task 2 — Motherboard Ports & Socket Identification

Label the actual motherboard with the following where present: SATA ports for SATA storage; motherboard 24-pin power connector; CPU power connector; USB headers; front-panel headers; fan headers; audio header; PCIe expansion slots; M.2 slots; and rear I/O ports such as USB, Ethernet, audio and video outputs. Identify the CPU socket printed or documented by the manufacturer and record the CPU families supported by that specific motherboard. The exact socket and supported CPUs must be taken from the hardware/manual rather than guessed.

Task 3 — RAM & Storage Installation

1. Shut down the PC, unplug AC power and use ESD precautions. 2. Open the case and locate the RAM. Release the retaining clips, remove the module by its edges, and avoid touching the contacts. 3. Align the RAM notch with the slot key and press evenly until the retaining clips lock. 4. Identify the storage connection as SATA or NVMe. 5. For SATA, disconnect/reconnect the data and power cables; for M.2/NVMe, remove the retaining screw, lift/remove the drive, reinstall it at the correct angle and secure it. 6. Reconnect power, boot into BIOS/UEFI and confirm RAM capacity and the storage device are detected.

Task 4 — Full Desktop PC Assembly

Recommended sequence: prepare the case and motherboard; install the CPU; install the CPU cooler and connect its fan; install RAM; install M.2 storage if applicable; fit the motherboard into the case with correct standoffs; connect motherboard and CPU power; install the GPU and connect its required power; connect SATA storage and power where applicable; connect front-panel, USB and audio headers; arrange cables safely; inspect for loose screws or disconnected cables; then power on and confirm POST. Document each stage with your own labeled photos or diagrams.

Section C — Mini Capstone Project

Question C1

Question: Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

Answer: First check the PSU and its power connections because the system cannot start correctly without stable power. Second check RAM because incorrectly seated or incompatible RAM can prevent POST. Third check the motherboard/CPU installation and CPU power connection because socket, power or compatibility problems can stop the system before normal startup. Motherboard diagnostic LEDs/beeps should be used to prioritize the next check.

Question C2

Question: Explain how motherboard–CPU compatibility affects system functionality.

Answer: The CPU must match the motherboard's physical socket and supported CPU family. The motherboard chipset and BIOS/UEFI version must also support that CPU. If the CPU is unsupported, incorrectly installed or lacks required power, the system may fail to POST or may not identify the processor correctly. For the specific assembled system, verify the socket, CPU support list and BIOS requirement using the motherboard manufacturer's documentation.

Question C3

Question: Describe how you would verify RAM and storage are installed correctly, including what BIOS/POST should confirm.

Answer: Power off and use ESD precautions. Reseat RAM so the notch aligns with the slot and the retaining clips lock. For storage, verify that SATA data and power cables are connected, or that an M.2 drive is correctly inserted and secured. During POST/BIOS, confirm the expected RAM capacity is detected and the storage drive appears in the storage/device information. If either is missing, power down and recheck seating, cables, slot choice and compatibility.

Question C4

Question: Produce a hardware-installation checklist of at least three steps to prevent similar failures.

Answer: 1. Verify compatibility before assembly: CPU socket/BIOS, RAM type, storage interface, PSU capacity/connectors and physical fit. 2. Use ESD precautions and follow the component/motherboard manuals during installation. 3. Before first power-on, inspect and firmly seat motherboard power, CPU power, RAM, GPU and storage connections. 4. Enter BIOS/UEFI after first boot and verify CPU, RAM and storage detection. 5. Document the build and any diagnostic indicators for future troubleshooting.

Question C5

Question: Provide a technical justification for using SSDs instead of HDDs in modern office workstations.

Answer: SSDs use flash memory and have no spinning platters or mechanical heads, so they provide lower access latency, faster boot and application loading, quieter operation and better resistance to physical movement. These characteristics fit office workstations where users frequently open applications and files. An NVMe SSD can provide additional performance when the platform supports it. HDDs can still be useful where large, low-cost capacity is the priority, but an SSD is the better primary system drive for the stated office workload.

Section D — AI-Augmented Learning

Step 1 — AI-assisted troubleshooting checklist

1. Power off and unplug the system; apply ESD precautions before touching internal components.
2. Check PSU switch, AC connection, PSU-to-motherboard 24-pin connection and CPU power connection.
3. Check motherboard diagnostic LEDs/beeps and consult the motherboard manual.

4. Remove and reseat RAM; if needed, test with the minimum supported RAM configuration recommended by the board manual.
5. Check CPU seating, cooler installation and CPU power connection; confirm CPU/motherboard compatibility and BIOS support.
6. Check GPU seating and required GPU power if the system uses a discrete GPU.
7. Check storage seating and SATA/NVMe connections. Confirm storage detection in BIOS/UEFI after POST is restored.
8. Boot to BIOS/UEFI and verify CPU, RAM and storage detection before proceeding to the operating system.
9. Relate the diagnosis to the ABC Solutions issue: the first failed physical/compatibility check that explains the symptom should be recorded as the likely cause and verified by reseating or correcting the component.

Step 2 — Test & Debug: corrected version

A safe correction is to make ESD precautions and power isolation the first step, before any internal inspection. The checklist above also places physical power, RAM, CPU/motherboard and storage checks before software-level troubleshooting. This avoids wasting time checking software when the machine has not successfully completed POST.

3–4 line note for submission: The checklist was reviewed for safety and troubleshooting priority. ESD precautions and power isolation were kept at the beginning so internal components are handled safely. Physical connections and POST/BIOS checks are performed before software troubleshooting because a newly assembled PC must first establish basic hardware functionality. The final version therefore reduces the risk of unsafe handling and incorrect diagnosis.

Submission Reminder

The assessment specifically requires: (1) the exact prompt(s) given to the AI tool, (2) the AI's original checklist and your corrected version, and (3) a 3–4 line note explaining the changes. For practical Section B tasks, attach your own labeled photos/screenshots/diagrams and fill in actual make/model/specifications from the hardware you inspected.